

BNY MELLON – INTERNSHIP INTERVIEW EXPERIENCE

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CSE batch of 2026, CEG

About the company

BNY is a global financial services company headquartered in New York City. BNY oversees nearly **\$50 trillion** in assets for our clients – managing it, moving it, and keeping it safe. For **240 years**, they have been creating new solutions that benefit businesses, communities, and people everywhere. It is the world's largest custodian bank and securities services company. The bank's primary functions are managing and servicing the investments of institutions and high-net-worth individuals. Its two primary businesses are Investment Services and Investment Management. The bank's clients include 80 percent of Fortune 500 companies. Currently, the company has over 12,000 employees across our three locations: Mumbai, Pune, and Chennai

No of Rounds: 3 (Online Test + Technical interview + Technical & HR interview)

Stipend: 75,000 INR per month

Duration of internship: 8-10 weeks

Round 1 – Online Assessment

Coding Round: Conducted in Hackerrank Website

Test Duration: 120 mins

Camera On

1 easy, 2 medium and 1 hard question.

Only function not full code

Coding questions:

1. Balancing Elements

When an element is deleted from an array, the higher-indexed elements shift down one index to fill the gap, A “balancing element” is defined as an element that, when deleted from the array, results in the sum of the even-indexed elements being equal to the sum of the odd-indexed elements.

Determine how many balancing elements a given array contains.

Example:

Input:

n=5

arr = [5,5,2,8]

Output:

2

When the first or second 5 is deleted, the array becomes [5, 2, 5, 8]. The sum[even] = 5+5=10 and sum[odd]=-2+ 8=10. No other elements of the original array have that property. There are 2 balancing elements: arr[0] and arr[1].

2. Maximum Difference

Given a number of nodes and edges, construct an undirected graph, connected component of the graph is any group nodes, For each connected component, determine the difference between its maximum and minimum value. Return the maximum of all of these differences.

For example given a graph with 4 nodes, connected as follows: (1, 2, 3), and (4). The only connected component has a difference $3-1=2$

Input:

```
int g_node;  
int g_from[g_edges];  
int g_to[g_edges];
```

Output;

```
int maximum_absolute_diff
```

3. Paint the Ceiling

Given n, s0, k, b, m, a

$s[i] = ((k*s[i-1] \bmod m) + 1 + s[i-1])$ for $1 \leq i \leq n$

Example:

$s=[2,4,6]$

Take all combination of 2 elements with repetition and find how many $s1*s2 \leq a$.

4. Positive Negative

The traders of Hackerland have stock prediction model that they want to backtest. The predicted profit earned every month by a certain portfolio is given by an array pnl of n integers where a negative integer denotes loss.

The traders decide to iterate through the months from 1 to n, and if the pnl is positive, they add it to their profit and if the pnl is negative, they can either subtract it or skip it. The net profit must always be greater than or equal to 0.

Given the array pnl and an integer k, find the maximum total profit that can be earned by the portfolio after subtracting at least k losses. If it is not possible to take at least k losses, report -1 as the answer.

Example: Suppose $n=5$, $k=2$ and $pnl=[4, -3, 6, -2, -1]$

The optimal strategy is to add the profits 4 and 6, and the losses $[-2, -1]$ to get a net profit of 7 which is the maximum possible.

For me 1st question 15/15, 2nd question 15/15, 3rd question 5/11, 4th question – 10/15 test cases passed.

24 students were selected for the next round. I was one of them.

Round 2 – Technical Interview

The interview was conducted in person at CUIIC. It went for about 30-35 min.

The interviewer first went through my resume and asked about the Clubs mentioned in my resume. Then she asked about the Tower of Hanoi problem. I tried but she corrected it. I gave the correct solution and then she was satisfied.

The interviewer started to test me in Data Structures. She asked about stacks, queues, and Binary Search Trees(BST) and I explained using paper and pen. For BST, I started by explaining about trees, and binary trees, and then I explained about BST.

The interviewer gave one scenario. Consider your friend or neighbor asking you to develop the pickle e-commerce for their business. You have to develop the website including the backend and database. She asked what tech stacks are you going to use, and what is the website architecture and asked about website flow.

Fortunately, this website looked like my DBMS project. I explained all the things. She'd asked why did you use PostgreSQL instead of MySQL or any other tool. I said that I already implemented PostgreSQL in my project. Then she asked what if I searched manga instead of mango, and asked me how to implement this. I said if there is only Tamil language. I will store all Tamil equivalent words in the database. For all region words, I will use Google Translate API to convert region words to English. She asked what if I searched maavadu instead of manga, and I said that I would use a dictionary for it.

The interviewer gave a Palindrome problem(<https://www.geeksforgeeks.org/c-program-check-given-string-palindrome/>) and I said the brute force approach. She asked me to write code on paper. She did not tell to optimize the code but I said I have optimized approach which is using two-pointer without any extra space, we can solve problem.

She asked questions related to my certifications. She asked what is REST API and I was able to answer. She asked full form of REST and I said I forgot it. Then, She asked full form of API and I answered. She inquired about what is stateless and stateful. I could not answer it correctly.

Overall, the interview went very well and I was shortlisted for the next round.

10 students were selected for the next interview. I was fortunate to be one of them.

Round 3 – Technical + HR Interview

It took place in person at CUIIC. It went for about 35 minutes.

The interviewer asked my resume and asked me to introduce myself. Then he went through my resume thoroughly. He asked which platform did you use to solve the problems. I said I am using Leetcode. He asked why? I said that compare to other platforms, Leetcode's UI and UX is nice to solve the problems. He asked do you like UI? I said yes.

The interviewer asked me what is HTTP default port number. He explained how client-server communication works and then asked the same question what is the HTTP default port

number? I took some time to think about it. He asked don't you know? I said yes. He moved to the next question.

Which language you know, he asked. I said C++. He gave me the paper to write code. The question is to reverse the string without an inbuilt function. I wrote the code from scratch. Then he asked me to input the string from the user. I wrote the code.

The interviewer gave one scenario. If you collect the data from one HTML form like first name, Last name, address..etc., then it will go to the backend and finally, it will be stored in the Database. What if any of the fields are entered with a script file or any malicious link and if it is stored in the database, it will be dangerous? How do you protect DB from it?

I said that I would use Node.js for the backend. In the node, there is a library to format the data from the form and it will not accept the malicious link.

The interviewer asked me HR questions and behavioral questions. I answered each question with an example.

The interviewer asked me an aptitude question. 20 people shake hands with each other. How many handshakes will be there in total? He asked the final answer. I answered 190 handshakes.

Finally, the interviewer asked me, do you have any questions? I asked what tech stacks are used in summer internships.

Overall, 6 of us from CEG and MIT got selected for the summer internship program at BNY, and I was fortunate to be one among them.

If you have any further questions, I will be very happy to address it :)

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All the very best!